

Raman Saparkhan

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EDUCATION

- **Carnegie Mellon University** Doha, Qatar
 - *Bachelor of Science in Computer Science, Dean's List* Aug. 2022 – May 2026
 - **Relevant Coursework:** Distributed Systems, Cloud Computing, Deep Learning, Machine Learning, Software Engineering, Parallel & Sequential Data Structures and Algorithms, Linear Algebra, Multivariate Calculus.
 - **Technical Leadership:** Google Developer Student Club Lead, Team Lead at AWS College Contributor Initiative.

EXPERIENCE

- **Carnegie Mellon University, CS department** Doha, Qatar
 - *Teaching Assistant for **Distributed Systems** Course* August 2024 - Present
 - Assisted instructors in teaching material via office hours, 1:1 sessions, data analytics workshops, and mentor meetings.
 - Applied big data frameworks **Ray, MapReduce, Hadoop, Spark** to enhance distributed systems understanding.
 - Mentored 35 students throughout their first project, leading to special recognition from the instructors.
- **Singularity Connect** Remote
 - *AI Engineer* July 2024 - Present
 - Developed a Full-stack AI app for customer service automation applying **Langchain, n8n, React & Rest API**.
 - Designed a robust architecture for the Artificial Intelligence system using **LangGraph, Tavily, Redis & Sqlight**.
 - Integrated **ReAct model, knowledge graphs, & semantic search** improving accuracy in AI workflow by 20%.
 - Collaborated with cross-functional teams, including product managers, data scientists, and operations, for troubleshooting and to align AI-driven customer service and software solutions with business objectives and user needs.
- **Carnegie Mellon University, Research Department** Doha, Qatar
 - *ML Research Engineer* May 2024 - July 2024
 - Enhanced Statistical Arbitrage by integrating Retrieval Augmented Generation **RAG algorithms**, engineering **Neural Networks** with **Keras & PyTorch, fine-tuning LLM Llama 3-8B** for sentiment analysis, achieving **90%** in return.
 - Built a data processing pipeline for 300 stock tickers using machine learning, data mining and integrating **Bloomberg Terminal**, Refinitiv and WRDS, developing a database management system with 300,000 news and time-series samples.
 - Developed custom backtesting platform for algorithm design & optimization, enhancing server architecture reliability.
 - Applied **Advanced Statistics**, regression, and probability models for risk management, reaching **80% win rate** and 5% drawdown.
- **Amazon Web Services (AWS)** Seattle, WA, USA
 - *Software Engineer Contributor* Mar 2023 - May 2024
 - Contributed to OpenSearch **Python & Machine Learning libraries** enhancing data analytics & system stability.
 - Developed local APIs and CI/CD testing pipelines, improving data backup, and operational efficiency of the system.
 - Featured speaker at **OpenSearchCon 2023**, sharing insights into the open-source development for college contributors, increasing engagement.
- **Qatar Computational Research Institution** Doha, Qatar
 - *Software Engineer Intern* Dec 2022 - Mar 2023
 - Developed a cross-platform **Flutter** app for businesses to organize files using URLs, documents, tags & categories.
 - Engineered a login page refresh system and category system, resulting in app performance improvement by 12%.
 - Migrated the app to null-safety and conducted end-to-end tests on **Android Studio**, utilized debugging tools, version control, and agile development with Cloud service **Azure DevOps**, increasing stability by 20%.

PROJECTS

- **AI Algorithmic Trader** | *Python, PyTorch, Tensorflow, NLTK, Selenium, Pandas, Scikit-learn, Llama 3-8B*
 - Implemented ensemble ML & NLP models, achieving **80%** precision on predicting stock price action using econometrics.
 - Increased quality and size of data by **35%** by employing **Selenium** for direct information retrieval and data mining.
 - Trained Large Language Models applying distributed computing on **NVIDIA A100 GPUs** on Linux Ubuntu Servers.
- **FileStack - Distributed File System** | *Java, Java RMI, TCP/IP, dynamic proxy*
 - Engineered Distributed File System (**DFS**) using **Java RMI & RPC** to address storage systems scalability challenges.
 - Programmed concurrency, synchronization, multi-threading, & object-oriented design, boosting system reliability & server performance by 30%.
- **SQL Query Generator** | *React, Node.js, Express, CSS, ChatGPT API, JavaScript*
 - Built a full-stack app with **OpenAI API** to translate natural language into **SQL queries**, for non-technical users.
 - Developed the backend with **Node.js & Express** and a user-friendly **React** interface, focusing on server architecture.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, Dart, Rust, HTML, CSS, JavaScript, SQL
- **Tools:** Git, PyTorch, TensorFlow, Docker, Android Studio, HTTP, Pandas, OpenCV, Selenium, DBMS
- **Technologies:** Node.js, Flutter, Azure, AWS, React, Express, Scikit-learn, Redis, Kafka, Elasticsearch, Cassandra